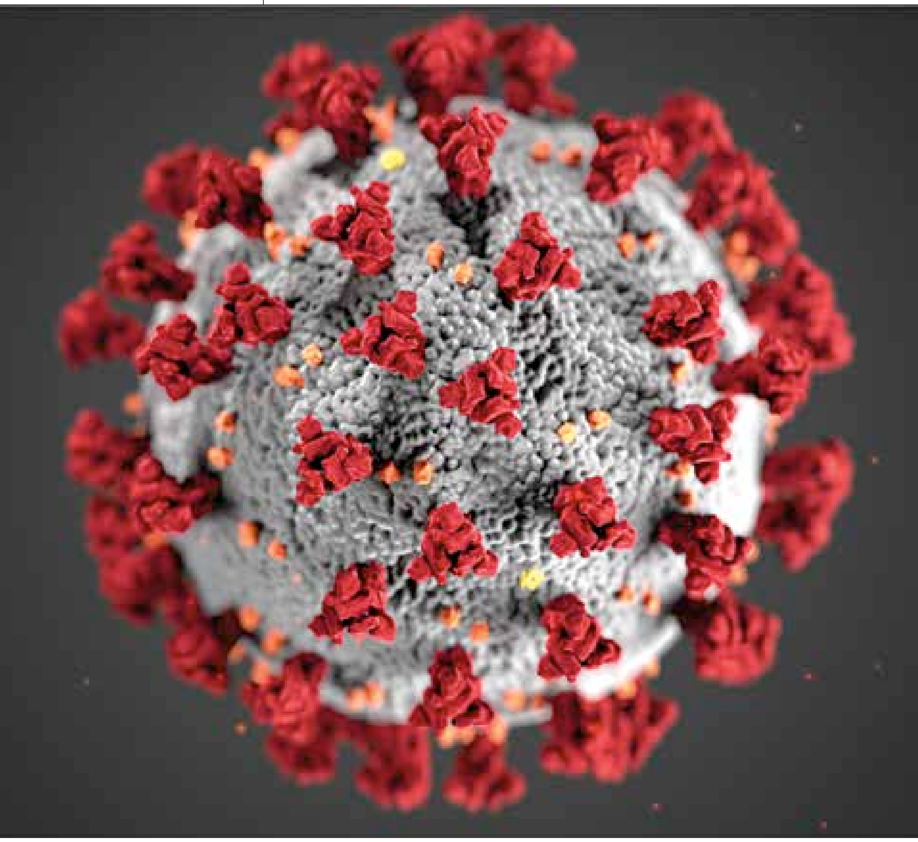
Protective measures
Here are some protective measures you can take to avoid contracting COVID-19.
https://dgit.in/may20-59

Q&A on COVID-19
WHO answers some commonly-asked questions about COVID-19 and protective measures. **https://dgit.in/may20-60**



Credit: Alissa Eckert, MS; Dan Higgins, MAMS

CORONAVIRUS OUTBREAKS

Despite decades of research, there are no vaccines or antiviral drugs that can prevent infections by coronaviruses

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In the 1950s, microbiologists were attempting to establish if any bacteria was the cause of the common cold, which was a frequent and seasonal infection suffered by humans worldwide. Once the researchers eliminated bacteria as a possible cause, the hunt began in earnest for the common cold virus. This search was rewarded, with the discovery of not one, but many types of viruses that all cause the common cold. The first of these to be identified was the rhinoviruses and adenoviruses in the 1950s, then the coronaviruses in the 1960s. All of these viruses were specific to humans. Coronaviruses affecting chicken and rats had been discovered earlier in the 30s and the 40s, although, at that time the researchers were not aware that the viruses affecting both the animals were related, and were part of the same family.

The rhinoviruses are responsible for half of all the common cold cases, while the coronaviruses cause about one fifth of the known cases. Scientists believe that there may be other viruses that cause the common cold that are yet to be isolated and identified. In the early days of the research, the relationship between the virus and the common cold was established by exposing volunteers to the virus, and then checking a box if they developed symptoms of the common cold. This was a rather primitive approach.

All these three families of viruses are constantly and rapidly evolving. Since their discovery, most of the research has been focused on the rhinoviruses, as they are the ones that are responsible for most of the common cold cases. For this reason, most of the research on antiviral drugs and treatment procedures by institutions and pharmaceutical companies were focused on rhinoviruses. The quest to cure the common cold has been going on for more than 60 years, but has, so far, been unsuccessful.

In the 1960s, virologists working on a number of different viruses including the ones that caused bronchitis in humans, hepatitis in mice, and gastroenteritis in swine found physical similarities in the transmission agents when imaged under an electron microscope. In these early electron microscopy scans, the club-shaped spikes that surround the virion appeared merged together in a smooth lighter region around a dark core. Considering the resemblance to the solar corona, these viruses were called coronaviruses.

In the 70s, there was a shift in how viruses were studied. Instead of focusing on how these disease causing agents affected the individual, and how the symptoms could